Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/26/2018
PI Name:	Behringer, Robert P. Ph.D.		
Project Title:	Compaction and Sound in Granular Matter		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Durham	**State:**	NC
Zip Code:	27708-0305	**Congressional District:**	1
Comments:	NOTE: PI Robert Behringer is deceased, as of July 2018.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	01/13/2015	**End Date:**	07/10/2018
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/10/2018 per passing of the PI (Ed., 12/4/18) NOTE: Original grant was NNX10AU01G, 10/1/2010-1/12/2015 (Ed., 12/15/17)		
Key Personnel Changes/Previous PI:	NOTE: Dr. Warren Warren, Duke University, has taken over the project due to the death of Dr. Behringer in July 2018 (Ed., 12/5/18)		
COI Name (Institution):			
Grant/Contract No.:	NNX15AD38G ; NNX10AU01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Original grant was NNX10AU01G, 10/1/2010-1/12/2015, entitled "Compaction and Sound Transmission in Dense Granular Media"; continuation grant established as "Compaction and Sound in Granular Matter," NNX15AD38G, 1/13/2015-1/12/2020. (Ed., 12/15/17) This project consists of a collaboration between researchers in France, Germany, and the United States, involving several leading teams in the field of granular materials/soft matter coming from these countries. The research program will exploit reduced-gravity conditions in order to understand the states of granular matter near jamming. Microgravity conditions are essential for achieving uniform granular samples, particularly near the jamming transition. Key scientific goals include: 1) Understanding the relation between classical ideas, including random-close and random-loose densities, and recent advances such as shear jamming, isotropic jamming, etc.; 2) A determination of critical properties of packings near jamming; 3) An investigation of sound transmission for packings near jamming. These studies will exploit both 2D and 3D techniques. The 3D techniques will exploit both photoelasticity and laser scanning devices. Two-dimensional photoelastic techniques have been extensively developed by the US Principal Investigator (PI), Robert Behringer, and his contribution will involve both quasi-static and acoustic studies of granular materials made from quasi-2D photoelastic particles. In addition, the US part of the collaboration will develop and use 3D systems based on laser-scanning techniques. Several technical goals for this project include: 1) further development of 3D scanning and reconstruction techniques; 2) further development of facilities for studying granular systems onboard the International Space Station (ISS), on sounding rockets, and on parabolic flights; 3) extension of highly quantitative photoelastic techniques from 2D to 3D; 4) the combining of photoelastic approaches and conventional acoustic measurement techniques; 5) the development of super-low friction substrates for acoustic measurements in 2D; 6) continued investigations that seek to understand jamming in 2D systems, where gravitational effect can be mitigated.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It has been known for a long time that stresses in granular materials stored in a container do not develop in the same way as the hydrostatic stresses in a fluid stored in the same container. Understanding the stresses in static, and slowly-moving, and dynamic dense granular materials is a research area of considerable importance, impacting such industries as pharmaceutical, cement, mining, chemical, and silo design/construction.
	Summary of Research Accomplishments (Covers years one to three) Publication 15 describes what happens to collections of grains, some of which contain embedded magnets. The presence of the grains can significantly change the nature of the jamming transition and the associated force networks. In fact, the high sensitivity of the jamming transition to additional forces is exactly the issue that arises for near-jamming systems subject to even weak gravitational forcing. This paper was a featured article in Europhysics Letters in 2016. Publication 17 seeks to understand how particle shape affects the nature of flow and clogging in a simple hopper geometry. In particular, we ask whether classical flow laws still apply, and what geometric effects occur during a clog that stops the flow. This is relevant to practical flow problems that might occur in industry, and also in NASA related situations, such as granular handling on the surface of the Moon. Year 3 results on jamming We have recently extended studies of star-shaped 3D particles to their 2D analogue, namely crosses, as discussed in publication 24. These results are remarkable for the very low densities at jamming, and for the novel orientational effects associated with the rotational symmetry of crosses. Paper 25 describes initial investigations of jamming for nonspherical (in 2D--hence, non-circular) particles in the shape of regular N-sided particles. In ongoing work, we are investigating a number of different measures that are used to determine jamming and near-jamming properties. In Paper 26, we pose the question, what is the response to a granular material as an embedded intruder is extracted. What are the networks that form and what are their properties? This project combines features of jamming, of impacts, and of a classical phenomenon known as the Janssen effect, involving the interaction of a gravitationally loaded system with its boundaries. Like paper 25, paper 27 explores the nature of jamming when the particles are non-spherical--in this case, ellipses. Note that a very interesting property of this system is the local ordering associated with the net alignment of the ellipses in some region Shear jamming occurs when a stress-free initial state is subject to shear strain, leading to a jammed anisotropic state, as first reported by Bi et al. (Nature. 2011 Dec 14;480(7377):355-8. doi: 10.1038/nature10667). Shear jamming occurs in a range of densities from random loose packing (rlp) to random close packing (rcp). In order to access shear jammed states at the lowest range of densities, it is necessary to apply large amounts of shear without generating a shear band. In earlier work, we achieved strains of a few 10's of percent without shear band formation by means of a novel simple shear experiment. The base of this system deforms uniformly (affinely) with the walls, applying uniform strain throughout. To reach the lowest shear jammed states, it is important to apply much larger strains, which we now can achieve with a novel ring-shear Couette experiment. In this experiment, the particles rest lightly on rings that can be rotated at individual rates, thus providing very large amounts of shear strain. Particle sizes are comparable to the rings. Paper 28 is a first exploration of the low density shear jammed states with this apparatus. Work is continuing on this project. Papers 29 and 30 explore structure and stability of packings of siz-armed star-shaped particles. These packings are very stable, and actually exhibit collective or geometric cohesion. Paper 35 investigates what local structures occur to enable shear jamming. We identify triplets of particles whose response leads to the formation of strong force chains, hence shear jamming. Paper 36 explores the network properties of force chains in experiments that characterize the onset of shear jamming. The goal is to relate very simple topological measures, such as the number connected segments and the number of `holes' to more conventional macroscopic properties such as the number of contacts per particle or the stresses. This study is undertaken in parallel with simulations by Lou Kondic and Lenka Kovalcinova at New Jersey Institute of Technology (NJIT) who are trying to capture and better understand the structural properties of shear jamming seen in

experiments.

Ongoing work on jamming

An important issue concerns whether 3D systems, particularly when the particles have low friction, exhibit the phenomenon of shear jamming, whereby a stress-free system can be caused to jam or become rigid by simply applying shear strain. Our 3D hydrogel systems provide an excellent setting for testing the hypothesis that low-friction 3D systems shear jam. To that end, we have built a new 3D shearing apparatus, which uses low-friction hydrogel particles and an improved laser scanning approach. Initial experiments using hydrogel particles suggest that shear jamming may occur. However, it turns out that for the very low-friction hydrogel particles that we have used to date, even a very low amount of gravity causes the system to jam, in the absence of compression or shear. Specifically, our imaging technique involves placing the hydrogel particles in an index-matched fluid, which has a density that is very close to but not precisely the same as that of the particles. Thus, our system has an effective gravity of only 0.01 g. Yet, that seems sufficient to spontaneously jam the system. To better understand the intertwined roles of gravity and particle friction, we have developed collaborations with simulation experts Lou Kondic and Lenka Kovalcinova at NJIT. For a particle friction coefficient of mu = 0:5, they have shown that 3D shear jamming does occur in the absence of gravity. We are currently preparing experiments with a new type of particles that have mu approximately equal to 0.2, where we expect to be able to disentangle friction and gravity in experiments.

High Speed Impacts into Granular Materials

A second class of studies involves high speed impacts of intruders into granular media. This work is presented in Year-1 publications 8, year-2 publication 13, and year-3 publications 22, 23, 26, 32-34, and 37. Understanding impacts is of considerable relevance to NASA in the context of meteorites impacting the Moon or other bodies.

Year 1 results on impacts

Publication 8 is a broad review of recent work carried out in the Duke lab, in collaboration with simulation expert, Lou Kondic from NJIT, on impacts into granular materials. In particular, we show that impacts are dominated by collisions of the intruder with latent force networks; momentum transfer along these networks provides the mechanism for inertial drag on the intruder. We have developed a collisional model for high speed impacts based on our 2D observations. This model successfully describes impacts for objects with a range of shapes.

Year 2 results on impacts

Publication 13 presents recent results from experiments and simulations where we consider the particle flow dynamics during impact into 2D granular materials, and show that the flow is slaved to the speed of the intruder. We relate this result to separation of time scales between inter-particle interactions and the intruder motion.

Year 3 results on impacts

We then extended our 2D tests to fully 3D impact studies to see if similar models would still work. In publications 22 and 23 we show that this is the case. Publication 26 turns the impact process upside down, in a sense, and seeks to understand granular dynamics that occur when a buried object, an intruder, is pulled out of a granular material. The protocol for this experiment consists of gradually increasing a vertically upward force on the intruder, until at some point the material fails and the intruder moves upward very rapidly. Below failure, the force networks, which reach to the vertical boundaries of the system, adjust to changes in the extracting force. After failure is triggered, the dynamics are more reminiscent of impacts, only turned upside down. We have also investigated failure behavior in a stick-slip failure experiment, in which a rough gravitationally loaded slider is pulled horizontally across a bed consisting of photoelastic disks. We present results from these experiments in publication 31, and ongoing work in this direction is designed to understand a transition from irregular stick-slip to periodic sliding as the pulling speed is increased. We also explore avalanche behavior in publication 33, where the avalanches are generated by cyclic shearing, and we explore creeping motion under cyclic shear in publication 33.

In a parallel set of studies, we have probed the nature of high speed intruder impacts into suspensions of cornstarch and dense fluids, such as water or ethanol. An object that is placed on the surface of such a suspension well sink, as long as the cornstarch fraction is not too high. In this same density regime, an intruder can be rapidly decelerated to zero velocity on impact, and may even rebound, before sinking slowly into the suspension. This process is known as impact activated solidification (IAS). This novel result is epitomized by u-tube videos of people running, without sinking, across pools of suspension. In publication 32, we carried out experiments where we were able to determine speeds associated with different transport processes, and to rule out one model for the hardening of cornstarch under impacts. In publication 37, we showed that cornstarch suspensions in water are quite different from those in ethanol, implying that part of the IAS process involves chemical processes with water and cornstarch.

Network Properties of Near-Jamming Granular Systems

A third class of studies involves using network analysis to characterize and understand granular states and their responses to applied protocols, such as compression, shear, etc. This work has appeared in Year-1 publications 1, 9-11, Year-2 publication 16, and Year-3 publication 36. Publications 1 and 9 focus on small scale topological structures that evolve under applied strains, such as shear, whereas 10 and 11 focus on larger scale network properties. Publication 16 provides a simple approach to characterizing vortices in granular shear flows. Publication 36 uses techniques from computational homology to provide a minimal description of shear jamming.

Reviews

Task Progress:

Publications 19 and 20 are invited reviews on the indicated topics. These are chapters of a book that appeared in 2016. Publication 21 is an invited paper that is a tribute to Prof. Horst Meyer who died in 2016. It describes how training in low temperature physics provided a background that was useful in developing completely new approaches for the study of granular materials. Publication 38 is an invited review that describes the nature of jamming in systems of grains with friction. Publications 1 and 2 are from the preceding grant, but connect to the current project as well. [Ed. note: preceding grant was NNX10AU01G, 10/1/2010-1/12/2015, entitled "Compaction and Sound Transmission in Dense Granular Media"]

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ED. NOTE (December 2018): Dr. Warren Warren, Duke University, has taken over the project due to the death of Dr. Behringer in July 2018 ; please see the record for this project under Dr. Warren's name for subsequent reporting.

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